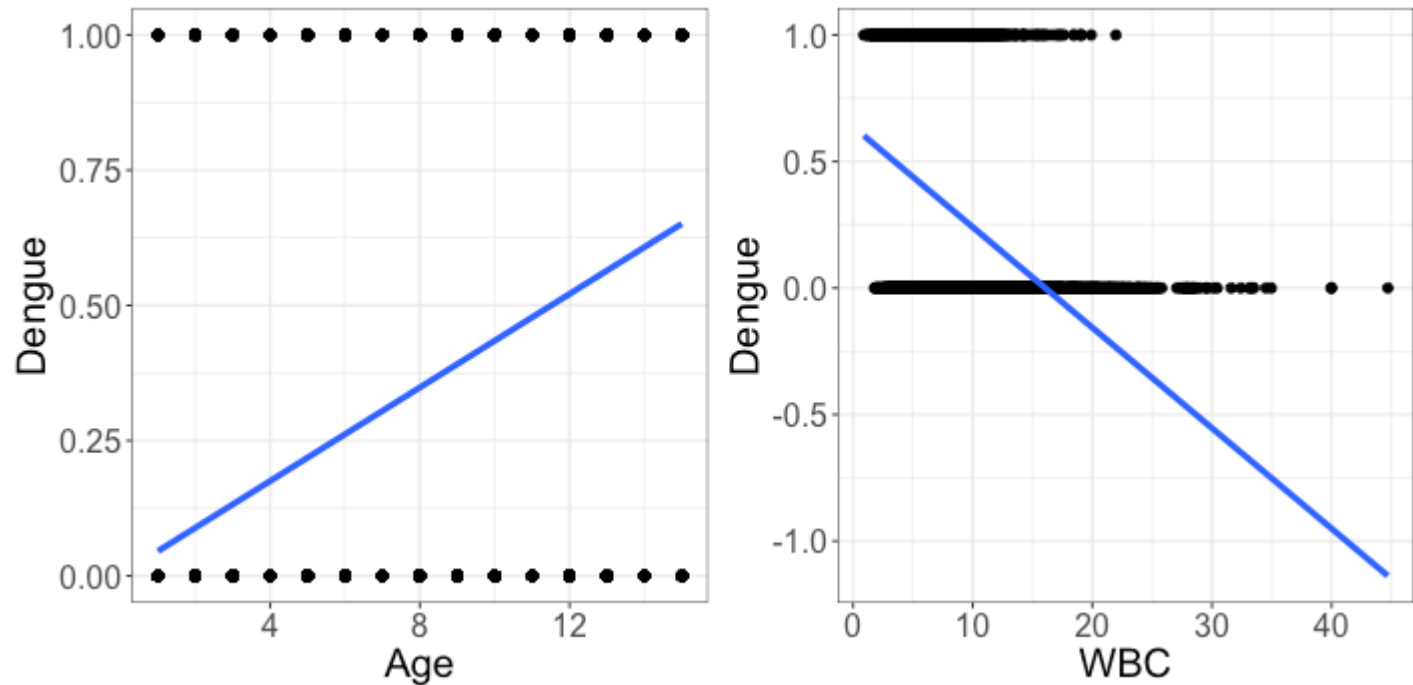


Parametric models and logistic regression

Ciaran Evans

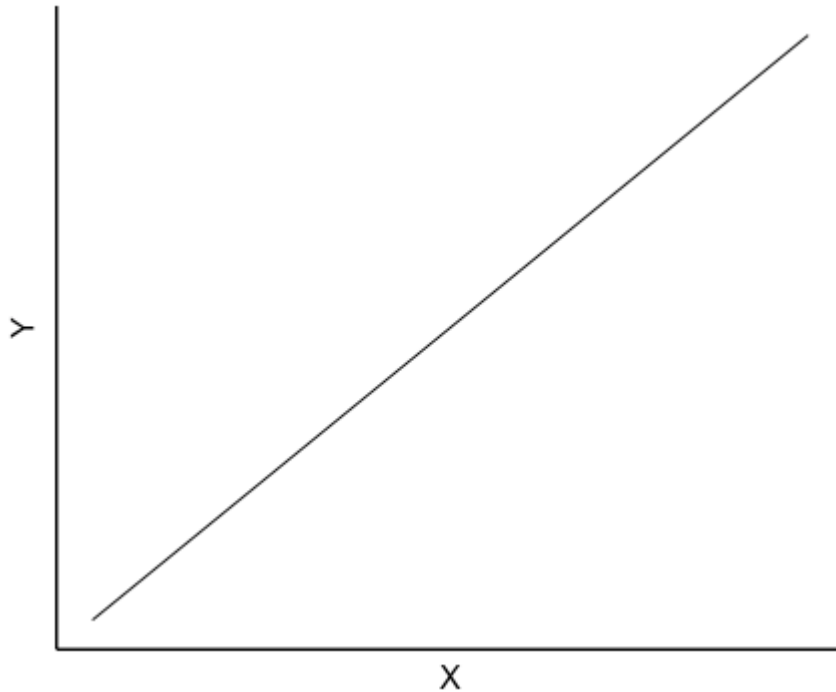
Last time

Don't fit linear regression with a binary response



Revisiting the linear regression model

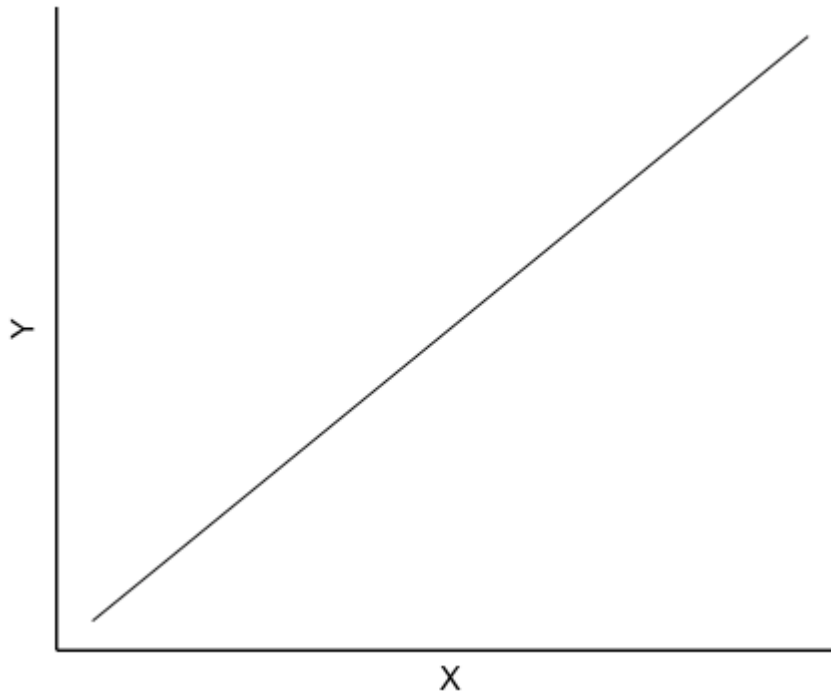
$$Y_i = \beta_0 + \beta_1 X_i + \varepsilon_i$$



Will all of the observations fall exactly on the line?

Revisiting the linear regression model

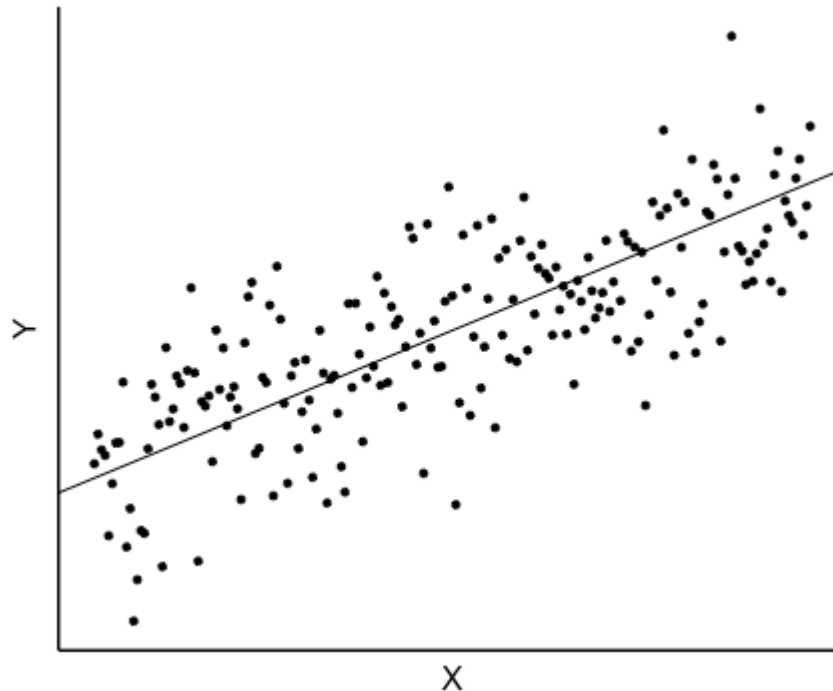
$$Y_i = \beta_0 + \beta_1 X_i + \varepsilon_i$$



Given a value of X , how do I know where the values of Y are likely to be?

Revisiting the linear regression model

$$Y_i = \beta_0 + \beta_1 X_i + \varepsilon_i$$



What do we often assume about the distribution of ε ?

Revisiting the linear regression model

Parametric modeling

A regression model is an example of a more general process called **parametric modeling**

- + **Step 1:** Choose a reasonable distribution for Y_i
- + **Step 2:** Build a model for the parameters of interest
- + **Step 3:** Fit the model

Step 1: Choose a reasonable distribution for Y_i

What do I mean by a *distribution*?

Step 1: Choose a reasonable distribution for Y_i

What do I mean by a *distribution*?

- + A **distribution** tells us what outcomes are possible for Y_i , and how often these outcomes occur.

Here the possible values of Y_i are 0 (no dengue) and 1 (dengue).

How often do these values occur in the population?

Step 1: Choose a reasonable distribution for Y_i

What do I mean by a *distribution*?

- + A **distribution** tells us what outcomes are possible for Y_i , and how often these outcomes occur.

Here the possible values of Y_i are 0 (no dengue) and 1 (dengue).

How often do these values occur in the population?

- + We don't know, so we will estimate from the sample
- + We assume the probability $Y_i = 1$ depends on Age_i

Step 1: Choose a reasonable distribution for Y_i

How should I describe the distribution of Y_i ?

Bernoulli distribution

Definition: Let Y_i be a binary random variable, and $\pi_i = P(Y_i = 1)$. Then $Y_i \sim \text{Bernoulli}(\pi_i)$.

What do I mean by a *random variable*?

Bernoulli distribution

Definition: Let Y_i be a binary random variable, and $\pi_i = P(Y_i = 1)$. Then $Y_i \sim \text{Bernoulli}(\pi_i)$.

What do I mean by a *random variable*?

A **random variable** is an event that has a set of possible outcomes, but we don't know which one will occur

- + Here $Y_i = 0$ or 1
- + Our goal is to use the observed data to estimate $\pi_i = P(Y_i = 1)$

Second attempt at a model

$$Y_i \sim \text{Bernoulli}(\pi_i) \quad \pi_i = P(Y_i = 1 | \text{Age}_i)$$

$$\pi_i = \beta_0 + \beta_1 \text{Age}_i$$

Are there still any potential issues with this approach?

Fixing the issues

Logistic regression model

Y_i = dengue status (0 = negative, 1 = positive)

Age_i = age (in years)

Random component: $Y_i \sim \text{Bernoulli}(\pi_i)$

Systematic component: $\log\left(\frac{\pi_i}{1 - \pi_i}\right) = \beta_0 + \beta_1 Age_i$

Logistic regression model

Y_i = dengue status (0 = negative, 1 = positive)

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Why is there no noise term ε_i in the logistic regression model?
Discuss for 1--2 minutes with your neighbor, then we will discuss as a class.

Fitting the logistic regression model

$$Y_i \sim \text{Bernoulli}(\pi_i)$$

$$\log\left(\frac{\pi_i}{1 - \pi_i}\right) = \beta_0 + \beta_1 \text{Age}_i$$

```
m1 <- glm(Dengue ~ Age, data = dengue,  
          family = binomial)  
summary(m1)
```

Fitting the logistic regression model

$$Y_i \sim \text{Bernoulli}(\pi_i)$$

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```
m1 <- glm(Dengue ~ Age, data = dengue,  
          family = binomial)  
summary(m1)
```

...

Coefficients:

##		Estimate	Std. Error	z value	Pr(> z)	
##	(Intercept)	-2.454345	0.075068	-32.70	<2e-16	***
##	Age	0.217312	0.008826	24.62	<2e-16	***

...

Class activity

- + Work with a neighbor on the class activity (handout)
- + I will collect your work at the end of class

For next time, read sections 6.4 and 6.6 in the textbook

Activity

$$\log\left(\frac{\hat{\pi}_i}{1 - \hat{\pi}_i}\right) = -2.45 + 0.22 \text{ Age}_i$$

Does the probability of dengue increase or decrease with age?

Activity

$$\log\left(\frac{\hat{\pi}_i}{1 - \hat{\pi}_i}\right) = -2.45 + 0.22 \text{ Age}_i$$

What are the predicted odds of dengue for a 5 year old patient?

Activity

$$\log\left(\frac{\hat{\pi}_i}{1 - \hat{\pi}_i}\right) = -2.45 + 0.22 \text{ Age}_i$$

For a 5 year old patient: is the predicted *probability* of dengue > 0.5 , < 0.5 , or $= 0.5$?